(c) no part of national output should be produced and consumed without passing through commercial channels

(d) a part of the national output being produced and consumed without passing through commercial channels is a sign of underdevelopment

**Ans. (b) :** In the passage, the author argues that in underdeveloped countries, a fraction of national output does not pass through commercial channels.
It means that it is not recognized explicitly in national income accounting or may ever go unrecognized.

## Passage 4

An increase in human-made carbon dioxide in the atmosphere could initiate a chain reaction between plant and microorganisms that would unsettle one of the largest carbon reservoirs on the planet soil. In a study, it was found that the soil, which contains twice the amount of carbon present in a plants and Earth's atmosphere combined, could become increasingly volatile as people add more carbon dioxide to the atmosphere. This is largely because of increased plant growth. Although a greenhouse gas and a pollutant, carbon dioxide also supports plant growth. As trees and other vegetation flourish in a carbon dioxide –rich future, their roots could stimulate microbial activity in soil carbon and its release into the atmosphere as carbon dioxide.

**67. Which among the following is the most logical corollary to the above passage ?**

(a) Carbon dioxide is essential for the survival of microorganisms and plants

(b) Humans are solely responsible for the release of carbon dioxide into the atmosphere

(c) Microorganisms and soil carbon are mainly responsible for the increased plant growth

(d) Increasing green cover could trigger the release of carbon trapped in soil

**Ans. (d) :** The most logical corollary to the above passage in that increasing green cover could trigger the release of carbon trapped in soil. On the one hand, while carbon dioxide is beneficial for plants, it is not beneficial for animals. At present, the amount of carbon dioxide in the earth is 0.03%.

## Passage 5

Historically, the biggest Challenge to world agriculture has been to achieve a balance between demand for and supply of food. At the level of individual countries, the demand-supply balance can be a critical issue for a closed economy, especially if it is a populous economy and its domestic agriculture is not growing sufficiently enough to ensure food supplies, on an enduring basis; it is not so much and not always, of a constraint for an open, and growing economy, which has adequate exchange surpluses to buy food abroad. For the world as a whole, supply-demand balance is always an inescapable prerequisite for warding off hunger a starvation. However, global availability of adequate supply does not necessarily mean that food would automatically move from countries of surplus to of deficit if the latter lack in purchasing power. The uneven distribution of hunger, starvation, under or mal-nourishment, etc., at the world –level, thus owes itself to the presence of empty-pocket hungry mouths, overwhelmingly confined to the underdeveloped economies. In as much as' a two square meal' is of elemental significance to basic human existence, the issue of worldwide supply of food has been gaining significance, in recent times, both because the quantum and the composition of demand has undergoing big changes, and because, in recent years, the capabilities of individual countries to generate uninterrupted chain of food supplies have come under strain. Food production marketing and prices, especially price-affordability by the poor in the developing world, have become global issues that need global thinking and global solutions.

**68. According to the above passage, which of the following are the fundamental solutions for the world food security problem ?**

1. Setting up more agro-based industries
2. Improving the price affordability by the poor
3. Regulating the conditions of marketing
4. Providing food subsidy to one and all

**Select the correct answer usig the code given below :**

(a) 1 and 2 only (b) 2 and 3 only
(c) 1, 3 and 4 only (d) 1, 2, 3 and 4

**Ans. (b) :** According to the above passage, there are two fundamental solutions for food security in the world-first to improve the price bearing capacity of the poor and secondly to provide regulation of marketing conditions.

**69. According to the above passage, the biggest challenge to world agriculture is :**

(a) to find sufficient land for agriculture and to expand food processing industries

(b) to eradicate hunger in uinderdeveloped countries

(c) to achieve a balance between the production of food and non-food items

(d) to achieve a balance between demand for and supply of food

**Ans. (d) :** According to the above passage, the biggest challenge before the world agriculture is to achieve a balance between the demand and supply of food.

**70. According to the above passage, which of the following helps/help in reducing hunger and starvation in the developing economies ?**

1. Balancing demand and supply of food
2. Increasing imports of food
3. Increasing purchasing power of the poor
4. Changing the food consumption patterns and practices

**Select the correct answer using the code given below :**

(a) 1 only (b) 2, 3 and 4 only
(c) 1 and 3 only (d) 1, 2, 3 and 4

**Ans. (c) :** According to the above passage, establishing a balance between the demand supply of food and increasing purchasing power of the poor helps in reducing hunger and starvation in the developing economies.



**71. The Issue of worldwide supply of food has gained importance mainly because of :**

1. overgrowth of population world wide
2. sharp declined in the area of food production
3. limitation in the capabilities for sustained supply of food.

**Select the correct answer using the code given below :**

(a) 1 and 2 only
(b) 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

**Ans. (b) :** The issue of worldwide food supply has gained importance due to the limitation in capacities for sustainable food supply.

**72. Four –digit numbers are to be formed using the digits 1,2,3 and 4; and none of these four digits are repeated in any mammer, Futher.**

1. 2 and 3 and are not to immediately follow each other
2. 1 is not to be immediately followed by 3
3. 4 is not to appear at he last place
4. 1 is not to appear at the first palce

**How many different numbers can be formed**

(a) 6
(b) 8
(c) 9
(d) None of the above

**Ans. (a) :** 6 different numbers can be formed i.e., (2143, 2431, 3142, 3412, 3421, 4312)

**73. A cylindrical overhead tank of radius 2 m and height 7 m is to be filled from an underground tank of size 5.5 m × 4m × 6cm. How much portion of the underground tanks is still filled with water after filling the overhead tank completely ?**

(a) 1/3 (b) 1/2
(c) 1/4 (d) 1/6

**Ans. (a) :**

The volume of cylindrical overhead tank = $\pi r^2 h$

$$V_1 = \frac{22}{7} \times 2 \times 2 \times = 7$$

$$V_1 = 88\ m^3$$

The volume of underground tank = $5.5 \times 4 \times 6$

$= V_2 = 132.0\ m^3$

The required portion of underground water

$$= \frac{132-88}{132} = \frac{44}{132} \Rightarrow \frac{1}{3}$$

after filling the over head tank.

**74. In a class of 60 students, where the number of girls is twice that of boys. Kamal, a boy, ranked seventeenth from the top. If there are 9 girls ahead of Kamal, the number of boys in rank after him is :**

(a) 13 (b) 12
(c) 7 (d) 3

**Ans. (b) :** Let the number of boys in the class = x

$\therefore$ The number of Girls 2x

According to question

$x + 2x = 60$

$3x = 60$

$x = 20$

$\therefore$ The number of boys (x) = 20

and the number of Girls (2x) = 40

The number of boys before Kamal = 16 – 9 = 7

The number of boys after Kamal = 20 – (7 + 1)

= 20 – 8 = 12

**75. A and B walk around a circular park. They start at 8 a.m. from the same point in the opposite directions. A and B walk at a speed of 2 rounds per hour and 3 rounds per hour respectively.**

**How many times shall they cross each other after 8.00 a.m. and before 9.30 a.m.?**

(a) 7 (b) 6
(c) 5 (d) 8

**Ans. (a) :**

Time taken by A to complete a round = $\frac{60}{2} = 30$ min.

and time taken by B to complete a round = $\frac{60}{3} = 20$ min.

$\therefore$ Relative speed of A and B = $\frac{1}{30} + \frac{1}{20}$

$$= \frac{2+3}{60} = \frac{5}{60} = \frac{1}{12}$$

Then time taken by A and B to complete the correct round = 12 min .

for the number of crossing each other,

$$\Rightarrow \frac{(9:30 - 8:00)}{12\,\text{min}}$$

$$= \frac{90\,\text{min}}{12\,\text{min}} = 7.5 \approx 7\,\text{times}$$

**76. W can do 25% of a work-in 30 days. X can do 1/4 of the work in 10 days, Y can do 40% of the work in 40 days and Z can do 1/3 of the work in 13 days . Who will complete the work first ?**

(a) W (b) X
(c) Y (d) Z